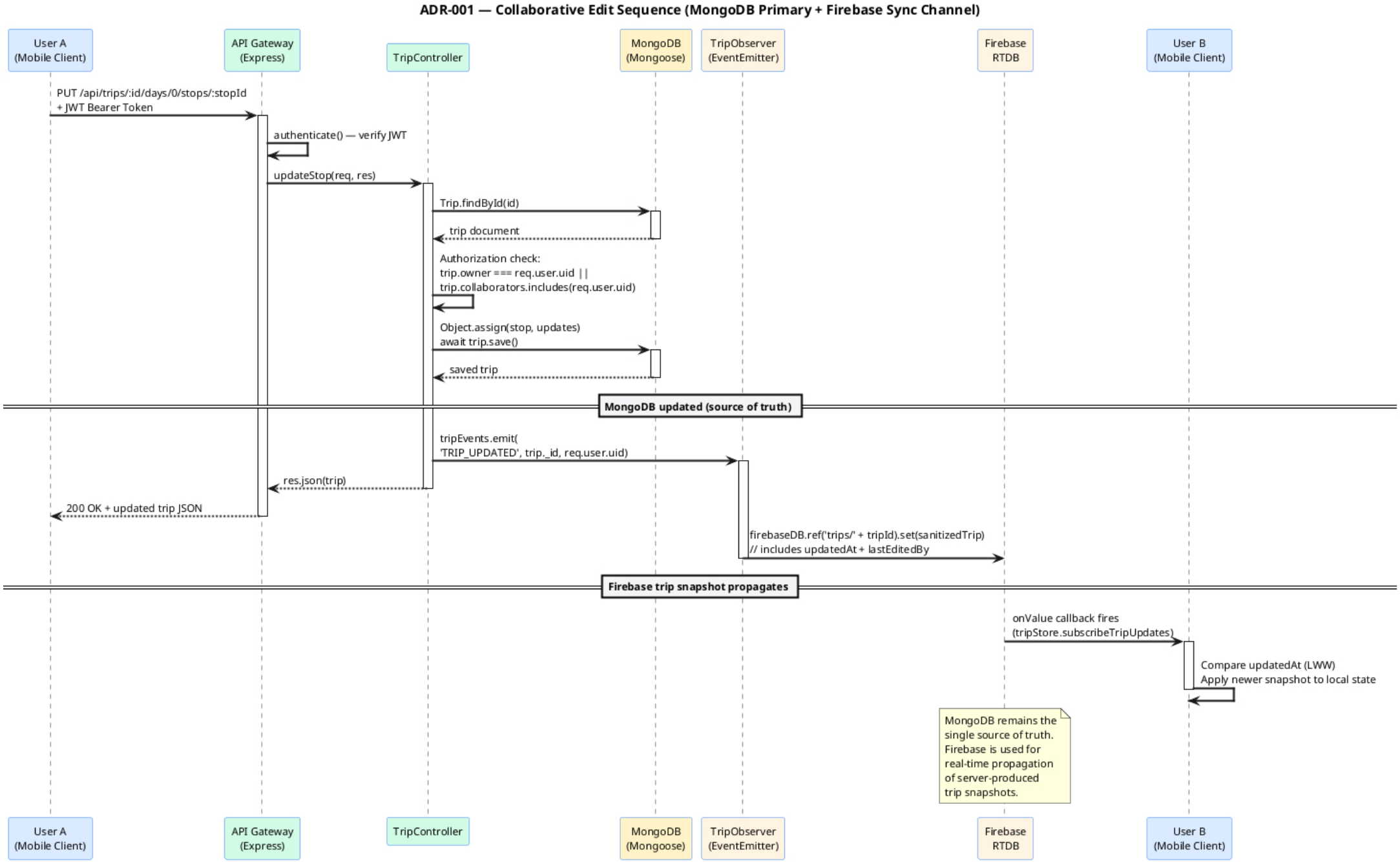
ADR-001 — Collaborative Edit Sequence (MongoDB Primary + Firebase Sync Channel)
User A (Mobile Client)
API Gateway (Express)
TripController
MongoDB (Mongoose)
TripObserver (EventEmitter)
Firebase RTDB
User B (Mobile Client)
PUT /api/trips/:id/days/0/stops/:stopId + JWT Bearer Token
authenticate() — verify JWT
updateStop(req, res)
Trip.findById(id)
trip document
Authorization check: trip.owner === req.user.uid || trip.collaborators.includes(req.user.uid)
Object.assign(stop, updates) await trip.save()
saved trip
MongoDB updated (source of truth)
tripEvents.emit('TRIP_UPDATED', trip._id, req.user.uid)
res.json(trip)
200 OK + updated trip JSON
firebaseDB.ref('trips/' + tripId).set(sanitizedTrip) // includes updatedAt + lastEditedBy
Firebase trip snapshot propagates
onValue callback fires (tripStore.subscribeTripUpdates)
Compare updatedAt (LWW) Apply newer snapshot to local state
MongoDB remains the single source of truth. Firebase is used for real-time propagation of server-produced trip snapshots.